Prodigy Anesthesia Quick Review

Safety Standards

What is the purpose of an isolation transformer in the operating room?	An isolation transformer isolates the operating room power supply from ground. Because most electrical shocks occur due to live current flow from the conductor of a grounded circuit through the person's body and back to ground, isolating the power supply from ground helps prevent this occurrence. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 17-18.*
What is the maximum leakage of electrical current allowed in operating room equipment?	The maximum leakage allowed in equipment used within operating rooms is 10 microamps. The leakage of an electrical current as low as 100 microamps can be fatal if the resistance of the skin is bypassed (as in surgical incision). *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 18.*
What governmental agency has authority over the safety standards for medical devices and what legislation gives them this authority?	In May 1976, the Medical Device Amendments gave the FDA specific authority for the regulation of medical devices. Within the FDA, device regulation falls under the auspices of the Center for Devices and Radiological Health. *Longnecker DE. Anesthesiology. New York: McGraw-Hill; 2008: 489.*
What is the primary cause of burns due to use of an electrosurgical unit?	The primary risk for burns from an electrosurgical unit are due to inadequate contact between the return pad and the skin or insufficient conductive gel. Dried gel on the patient surface of the contact pad reduces the area of contact with the patient as does placement of the pad over a bony prominence. It should be noted that some return pads do not use gel on the contact surface, but instead have a conductive adhesive to help ensure contact with the skin. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 19-20.*
Which are considered safer on the anesthesia machine, ascending or descending bellows?	Ascending (standing) bellows are considered safer than descending (also known as hanging) bellows. In the event of a circuit disconnect, ascending bellows will fail to rise whereas descending bellows will continue to descend due to gravity and ascend during the inspiratory phase. *Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 54.*
Why should you avoid pressing the oxygen flush valve on the anesthesia machine during inspiration?	Pressing the oxygen flush valve during inspiration can result in over-distention of the lungs and barotrauma because the flush valve delivers 35 to 75 L/min of oxygen at 50 psi. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 64.*

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Safety Standards

What anesthesia machine alarms are based on the circuit pressure?	The disconnect alarm and the high pressure alarm can be triggered by circuit pressure alarms. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 278-279.*
What anesthesia machine alarms are based upon the exhaled tidal volume?	Alarms based upon the exhaled tidal volume include low tidal volume, disconnect (which has multiple triggers), high respiratory rate, reverse flow, low minute volume, and ascending bellows. *Nagelhout JJ, Zaglaniczny KL. Nurse Anesthesia. 2nd ed. Philadelphia, PA: WB Saunders Company: 276.*

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